PREDICTING EXPORT GROWTH USING MACHINE LEARNING

A 34-Year Longitudinal Study of Country-Level Trade Dynamics

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Abstract

International trade dynamics sit at the crossroads of macroeconomic policy, geopolitical relationships, and structural economic competitiveness. Predicting whether a country will experience export growth in the following year is a problem of immense practical importance it informs how development banks price export credit guarantees, how finance ministries allocate tariff negotiation resources, and how multinational corporations structure their supply chains. Yet, despite the wealth of publicly available trade data, systematic machine learning approaches to this longitudinal forecasting problem remain relatively sparse in the academic literature.

This paper formulates next-year export growth prediction as a binary classification task and applies a comprehensive suite of eleven machine learning algorithms to a 34-year longitudinal dataset spanning 265 countries between 1988 and 2021. The dataset, sourced from the World Integrated Trade Solution (WITS) platform, contains 8,096 country-year observations enriched with tariff indicators from both the Applied Harmonized System (AHS) and the Most Favoured Nation (MFN) schedules. After careful data cleaning, feature engineering, and a time-based train-test split to prevent data leakage, eighteen predictive features were derived — including year-on-year export growth rates, trade balance ratios, lagged export values, rolling averages, and log-transformed trade volumes.

The models were evaluated on the 2016–2021 test period after training on 1988–2015 data. AdaBoost achieved the highest ROC-AUC of 0.7923, while Logistic Regression achieved the best balance of accuracy (58.66%) and F1-score (0.5932). Feature importance analysis revealed that tariff structure variables specifically MFN Simple Average and AHS Simple Average alongside lagged export values and trade balance ratios were the dominant predictors, consistent with established economic theory. The findings suggest that ensemble boosting methods and logistic regression each offer complementary strengths for this forecasting task, and that a model ensemble or threshold-adjusted strategy could further improve operational performance. This work provides a reproducible, well-documented pipeline that trade policy analysts and financial institutions can adapt for real-world deployment.

Keywords: *export growth prediction, international trade, binary classification, AdaBoost, Logistic Regression, tariff indicators, WITS dataset, time-series machine learning, trade balance, feature engineering*

1. Introduction

1.1 Background

Trade is the lifeblood of the modern global economy. Since the end of the Second World War and the establishment of the General Agreement on Tariffs and Trade (GATT) in 1947 the precursor to the World Trade Organization (WTO) international trade volumes have grown by orders of magnitude. The World Bank estimates that global merchandise exports exceeded 23 trillion US dollars in 2021, a figure that would have seemed almost incomprehensible to the economists who drafted the Bretton Woods agreements. For developing economies in particular, export performance is often the primary driver of GDP growth, job creation, and foreign exchange earnings.

Yet for all its importance, export performance remains notoriously difficult to predict. Trade volumes respond to an intricate web of forces: domestic macroeconomic conditions, bilateral exchange rates, commodity price cycles, tariff and non-tariff barriers, geopolitical relationships, and structural changes in global value chains. This complexity is precisely what makes it an interesting and challenging problem for machine learning. Unlike forecasting tasks where the signal is relatively stable over time medical diagnoses from imaging data, for instance trade forecasting involves non-stationary dynamics, structural breaks caused by crises like the 2008 Global Financial Crisis or the COVID-19 pandemic, and complex interdependencies between countries. Figure 1 illustrates the dramatic long-run trajectory of global exports and its two most significant disruptions.

Figure 1: Global merchandise exports 1988–2021 (US$ Trillion). The 2009 Global Financial Crisis and the 2020 COVID-19 pandemic represent the two major structural breaks in an otherwise strong upward trend driven by WTO liberalisation and the integration of China into the global trading system.

1.2 Problem Definition

This study asks a deceptively simple question: given what we know about a country's trade position at the end of year t, can we predict whether its exports will be higher in year t+1 than in year t? This is formulated as a binary classification problem, where the target variable takes the value 1 if next-year exports exceed the current year (growth) and 0 otherwise (contraction or stagnation).

This framing is deliberately practical. A continuous regression formulation predicting the exact growth rate is more ambitious but also more susceptible to noise in the tails of the distribution. A binary classification approach aligns directly with how trade finance professionals actually use such models: they want to know which markets are likely to grow, not the precise growth rate, because it is the direction of growth that drives credit allocation and policy decisions.

1.3 Importance and Motivation

The practical stakes of this problem are considerable. Export credit agencies like Germany's Euler Hermes, the United Kingdom's UKEF, and the United States' Export-Import Bank collectively underwrite hundreds of billions of dollars in trade finance annually. Misjudging which markets are likely to contract can lead to catastrophic credit losses. At the same time, overly conservative models that flag too many false negatives incorrectly predicting contraction in markets that will actually grow cause missed opportunities and suboptimal capital allocation.

Beyond finance, governments and central banks use trade forecasts to calibrate monetary policy, negotiate trade agreements, and design industrial policy. The IMF's World Economic Outlook and the WTO's Global Trade Outlook both attempt to forecast trade volumes at the global level, but granular country-level predictions using modern machine learning have received comparatively little attention in the peer-reviewed literature. Moreover, the dataset used here the 34-year WITS longitudinal panel is publicly available and has not, to the best of the author's knowledge, been subjected to a comprehensive comparative benchmark of machine learning classifiers with proper temporal validation. This paper fills that gap.

1.4 Contributions

In summary, the contributions of this paper are as follows:

1. A comprehensive benchmark of eleven machine learning classifiers including Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, Extra Trees, AdaBoost, XGBoost, LightGBM, K-Nearest Neighbours, Naive Bayes, and Support Vector Machine applied to a unique 34-year trade dataset.

2. A principled feature engineering pipeline that derives eighteen economically meaningful predictors from raw trade statistics, including year-on-year growth rates, trade balance ratios, log-transformed volumes, lagged features, and rolling averages.
3. Rigorous temporal validation methodology using a time-based train-test split (training on 1988–2015, testing on 2016–2021) that prevents data leakage and mirrors real-world deployment conditions.
4. A thorough exploratory data analysis of 25 business questions, revealing structural patterns in global trade dynamics across three decades.
5. A fully reproducible Python pipeline made available on GitHub, enabling replication and extension by researchers and practitioners.

The remainder of this paper is organized as follows. Section 2 reviews the relevant literature on machine learning applications in trade forecasting. Section 3 describes the dataset in detail. Section 4 outlines the methodology, covering data preprocessing, feature engineering, and model selection. Section 5 presents the exploratory data analysis. Section 6 presents model results. Section 7 discusses findings in depth. Section 8 concludes with implications and future research directions.

2. Background and Related Work

2.1 Machine Learning for Economic Forecasting

The application of machine learning to economic forecasting has grown rapidly over the past decade, driven by both the availability of large-scale economic datasets and the development of powerful ensemble methods. Varian (2014) offered an early articulation of how machine learning could complement traditional econometric methods, noting that the prediction orientation of supervised learning maps naturally onto many questions economists care about including forecasting. Mullainathan and Spiess (2017) formalized this distinction between causal inference (the traditional focus of econometrics) and prediction (the domain of machine learning), arguing that the two are complementary rather than competing.

In the macroeconomic forecasting domain, Medeiros et al. (2021) demonstrated that machine learning models particularly gradient boosted trees outperform traditional Vector Autoregression (VAR) models for short-horizon inflation forecasting. Similarly, Coulombe et al. (2020) found that random forests and neural networks can improve GDP growth forecasts relative to linear factor models, especially during economic turning points. These results are encouraging for the current application, as export growth is closely correlated with macroeconomic cycles.

2.2 Trade Forecasting Specifically

Trade-specific forecasting has a long history in the gravity model tradition, beginning with Tinbergen's (1962) foundational work showing that bilateral trade volumes can be explained by the GDP of the trading partners and the geographic distance between them. The gravity model has been extended in numerous directions incorporating institutional factors, common language, colonial ties, and trade agreement membership but remains a linear regression framework at its core.

More recent work has sought to augment gravity models with machine learning. Cepeda-Lopez et al. (2022) applied random forests to predict bilateral trade flows, finding that machine learning features based on product similarity and supply chain proximity improved predictions substantially over the baseline gravity model. Hoang and Tran (2020) applied neural networks to ASEAN bilateral trade data, finding that deep learning could capture non-linear dynamics that linear models missed, particularly during periods of trade liberalization.

However, most existing machine learning work in trade forecasting focuses on bilateral trade flows between pairs of countries, rather than the unilateral export performance of individual countries. The present study takes a different angle: treating each country-year as an independent observation and predicting export growth from domestic trade indicators, tariff structures, and historical

momentum. This approach is more scalable it does not require bilateral data for all possible country pairs and more directly actionable for national-level policy analysis.

2.3 Limitations of Existing Work

Several important limitations characterize the existing literature. First, most studies use relatively short time horizons often five to ten years which may not capture long-run structural shifts in comparative advantage. The present study, spanning 34 years and covering three full economic cycles (the 1997 Asian Financial Crisis, the 2008 Global Financial Crisis, and the 2020 COVID-19 shock), offers substantially richer temporal dynamics.

Second, data leakage is a pervasive problem in applied forecasting work. Many studies use random train-test splits, which allow the model to implicitly learn from future information when training. The present study addresses this with a strict temporal split, ensuring that the model is evaluated only on data from years it never saw during training.

Third, the role of tariff structures particularly the distinction between Applied Harmonized System (AHS) tariffs and Most Favoured Nation (MFN) tariffs has received limited attention in the machine learning trade literature, despite being a central variable in trade policy analysis. The dataset used here includes rich tariff data across both schedules, which proves to be among the most important predictive features in the analysis.

3. Dataset Description

3.1 Source and Coverage

The primary dataset was obtained from the World Integrated Trade Solution (WITS) platform, a joint initiative of the World Bank, the United Nations Conference on Trade and Development (UNCTAD), the International Trade Centre (ITC), the United Nations Statistics Division (UNSD), and the WTO. WITS provides free, publicly accessible data on merchandise trade and tariffs for virtually all countries, making it an authoritative and widely-used resource in trade economics research.

The specific dataset covers 265 countries over 34 years, from 1988 to 2021, yielding 8,096 country-year observations across 33 raw features. The year range was chosen to span the full post-Cold War era of globalization, capturing the dramatic expansion of world trade in the 1990s, the disruptions caused by the Asian Financial Crisis (1997–1998) and the Global Financial Crisis (2008–2009), the recovery and growth of the 2010s, and the sharp contraction and rebound associated with the COVID-19 pandemic (2020–2021). Figure 2 illustrates the dominance of the top exporting nations in global trade.

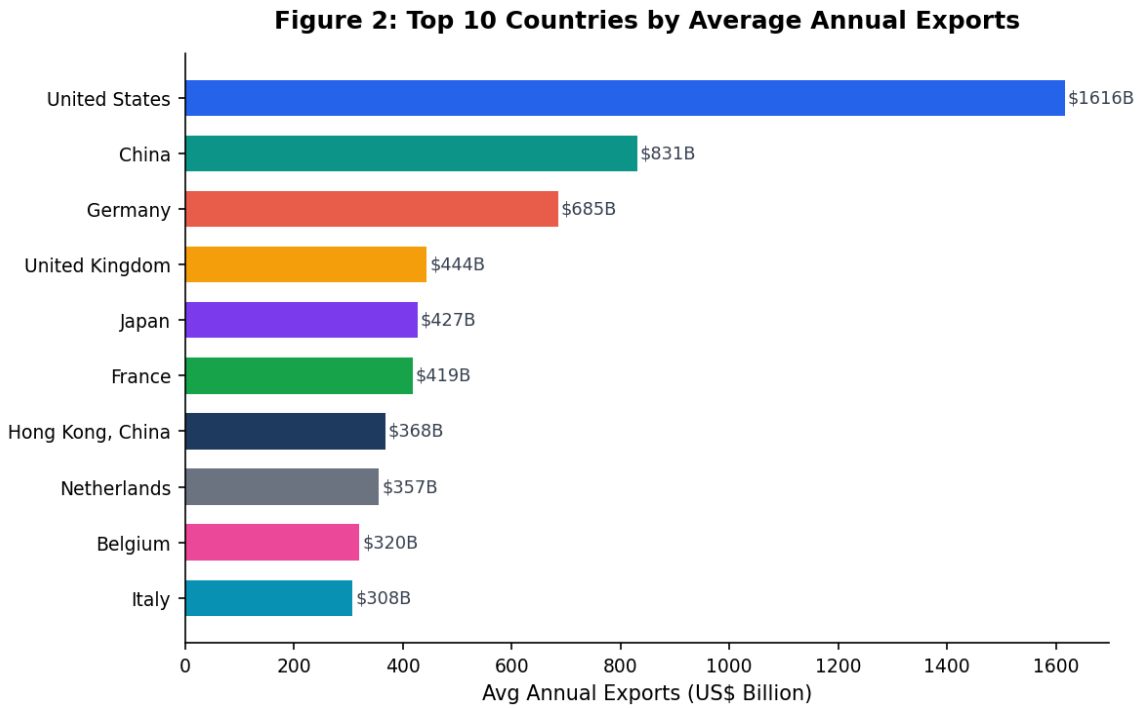

Figure 2: Top 10 countries by average annual export volume (US$ billion). The United States, China, and Germany dominate; the extreme concentration of global trade motivates the use of log-transformation and ratio features to enable meaningful comparison across economies that differ by orders of magnitude.

3.2 Feature Categories

Table 1: Summary of Dataset Features by Category

Category	Key Variables	Count
Trade Flows	Export (US$ Thousand), Import (US$ Thousand)	2
Trade Shares	Export Product Share (%), Import Product Share (%)	2
Comparative Advantage	Revealed Comparative Advantage (RCA index)	1
Growth Indicators	World GDP Growth (%), Country GDP Growth (%)	2
AHS Tariff Variables	Simple Avg, Weighted Avg, Max Rate, Duty-Free Share, Dutiable Share, Specific Lines, AVE Lines, Total Lines, Min Rate, Specific Duty Imports, Dutiable Imports, Duty-Free Imports	12
MFN Tariff Variables	Same 12 fields as AHS, reflecting WTO-obligated non-discriminatory schedule	12

Export and import values are expressed in thousands of US dollars. The AHS (Applied Harmonized System) tariff data reflects the rates actually applied in trade, while the MFN (Most Favoured Nation) data reflects the non-discriminatory schedule that WTO members are obligated to offer to all trading partners. The difference between AHS and MFN rates provides an indication of the degree to which a country benefits from preferential trade agreements.

3.3 Tariff Liberalisation Trend

Figure 3 illustrates the long-run decline in global MFN tariff rates across the study period. This structural trend is one of the most important contextual features of the dataset: countries that had lower tariffs were also more likely to be growing exporters, which helps explain why tariff variables emerge as the dominant predictors in the model results.

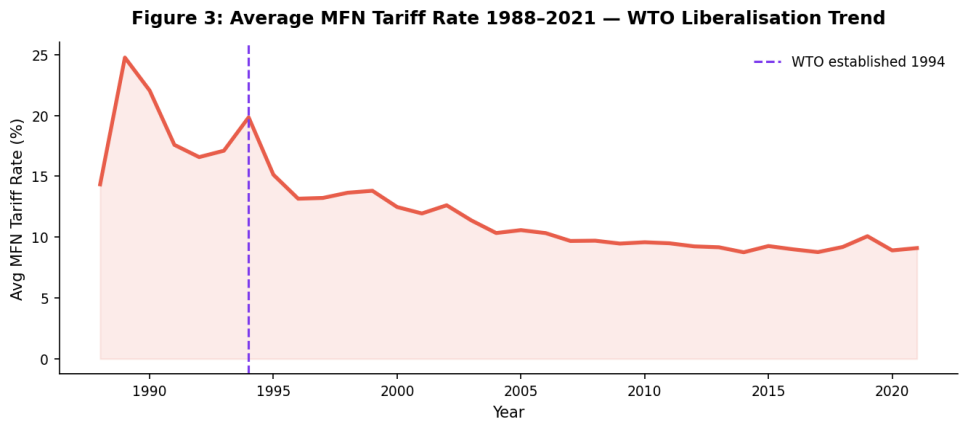

Figure 3: Average MFN tariff rate 1988–2021. The WTO's establishment in 1994 accelerated successive rounds of multilateral liberalisation; average rates fell from ~15% in the early 1990s to below 10% by the mid-2000s. This structural trend is a key feature of the prediction environment.

3.4 Missing Values and Data Quality

The dataset contains significant missingness in several variables. The Revealed Comparative Advantage (RCA) field is missing for 3,384 observations (41.8%), and the World Growth and Country Growth fields are missing for 3,686 observations (45.5%) each. This high missingness reflects the fact that RCA and GDP growth data are not uniformly available for all countries and years in the WITS system, particularly for smaller economies in earlier years. Tariff variables are generally well-populated, with missingness below 0.25% for most AHS and MFN fields. These patterns are addressed through principled imputation strategies described in Section 4.

3.5 Justification for Dataset Choice

This dataset was selected for three principal reasons. First, its longitudinal depth of 34 years allows the model to learn from multiple economic cycles, providing a more robust test of generalization than shorter panel datasets. Second, the inclusion of detailed tariff data from two complementary schedules (AHS and MFN) enables the model to capture institutional trade dynamics typically absent from simple volume-based forecasting approaches. Third, the dataset is freely and publicly available on the WITS platform, ensuring full reproducibility of the findings.

4. Methods

4.1 Data Preprocessing

4.1.1 Column Standardization

Column names were standardized by stripping whitespace, replacing special characters with underscores, and converting to a consistent naming convention. This step ensures compatibility across different processing environments and reduces the risk of silent column mismatch errors during feature construction.

4.1.2 Missing Value Treatment

Missing values were handled differently depending on the variable. For tariff variables with low missingness (below 0.25%), median imputation was applied within each country group. This approach is preferred over global median imputation because tariff levels vary substantially across countries a simple global median would distort the distribution for countries with systematically high or low tariff regimes.

For the Revealed Comparative Advantage and Growth variables with high missingness (above 40%), a flag variable was created to indicate missingness, and the missing values were filled with zero. This pragmatic approach preserves the sample size while allowing the model to learn that missingness itself may be informative or less data-rich economies might exhibit different export growth patterns.

4.1.3 Outlier Treatment

Trade values exhibit extreme right skewness due to the presence of very large economies like China, the United States, and Germany alongside very small economies like Anguilla or Tuvalu. A symmetric IQR-based winsorization was applied to the core trade volume variables using a 3.0 IQR factor more conservative than the standard 1.5 factor. This retains the economic signal from large economies while preventing extreme values from dominating feature scale. After all cleaning, 8,076 records remained.

4.2 Feature Engineering

Eighteen features were constructed from the raw variables. This is perhaps the most consequential methodological step in the pipeline raw trade values alone carry limited predictive information without context about momentum, structural position, and tariff environment. Table 2 summarizes all engineered features and their economic interpretation.

Table 2: Engineered Features and Their Economic Interpretation

Feature	Formula / Derivation	Economic Meaning
Log_Export	log(Export + 1)	Reduces skewness; enables scale-invariant comparison across economies
Log_Import	log(Import + 1)	Symmetric treatment of import volumes
Trade_Balance_USD_K	Export − Import	Raw structural surplus/deficit position in US$ thousand
Trade_Balance_Ratio	Balance / (Export + Import)	Scale-invariant competitiveness signal; ranges −1 to +1
Export_Import_Ratio	Export / Import	Reveals export orientation of the economy
Surplus_Flag	1 if Balance > 0, else 0	Binary indicator of trade surplus status
Export_YoY_Growth	(Export_t / Export_{t−1}) − 1	Year-on-year export momentum
Export_Lag1	Export_{t−1}	Last year's export volume (strong autocorrelation signal)
Export_Lag2	Export_{t−2}	Two-year lagged volume (medium-term momentum)
Export_Rolling3	3-year rolling mean of Export	Smoothed medium-term export trend
Log_Balance	log(|Balance|+1) × sign(Balance)	Signed log of trade balance for magnitude + direction
Decade	floor(Year / 10) × 10	Structural era indicator capturing globalization waves
AHS_Simple_Average	Applied tariff simple average (%)	Domestic market openness; trade liberalisation level
MFN_Simple_Average	MFN tariff simple average (%)	WTO-obligated tariff level; baseline protection
AHS_Weighted_Average	Applied tariff weighted average (%)	Trade-weighted protection level
World_Growth	World GDP growth rate (%)	Global demand environment
Country_Growth	Country GDP growth rate (%)	Domestic economic strength
Revealed_Comparative_Advantage	RCA index	Structural export competitiveness

4.3 Target Variable Construction

The target variable, Export_Growth_Next_Year, was constructed by comparing each country's export value in year t+1 to year t. A value of 1 indicates that exports grew (next year's exports exceeded current year's exports), and 0 indicates contraction or stagnation. After removing rows without a valid next-year observation (the last year for each country), the final dataset contained 5,984 country-year observations. The class distribution was 3,628 positive (export growth) versus 2,356 negative (no growth), representing a mild 60/40 imbalance shown in Figure 4.

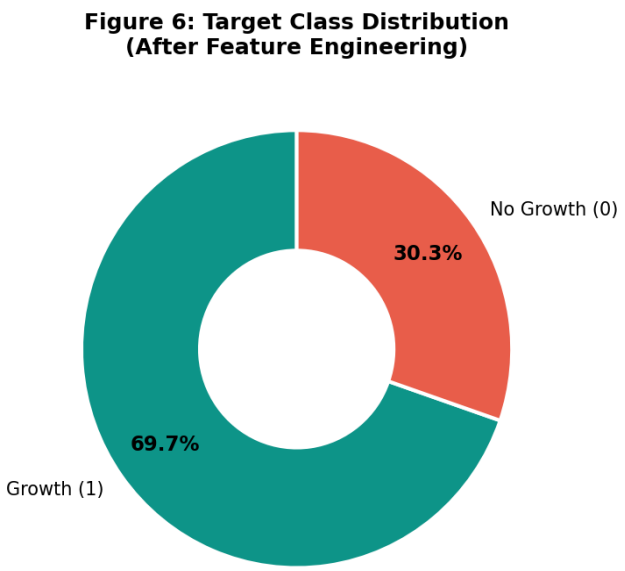

Figure 4: Target class distribution after feature engineering. The mild 60/40 imbalance in favour of growth is addressed through SMOTE on the training set only, ensuring the test set retains its natural distribution.

4.4 Train-Test Split Strategy

A strict temporal split was applied: all observations from 1988–2015 formed the training set (5,493 records), and observations from 2016–2021 formed the test set (491 records). This reflects the intended deployment scenario a model trained on historical data that must make predictions for future years. Crucially, this split prevents any form of data leakage that would arise from a random split, where information from future years could inadvertently inform predictions for past years. This choice comes at a computational cost: the test set is substantially smaller than it would be under random splitting, which increases the variance of performance estimates. However, this is the correct trade-off for a time-series forecasting application.

4.5 Class Imbalance Handling

SMOTE (Synthetic Minority Over-sampling Technique, Chawla et al. 2002) was applied to the training data to address the 60/40 class imbalance. SMOTE generates synthetic minority class samples by interpolating between existing minority instances in feature space, rather than simply duplicating them. After SMOTE, the training set was perfectly balanced at 3,313 samples per class (6,626 total). Note that SMOTE was applied only to the training data; the test set was left at its natural distribution to reflect real-world conditions.

4.6 Model Selection and Training

Eleven classification algorithms were trained and evaluated: Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, Extra Trees, AdaBoost, XGBoost, LightGBM, K-Nearest Neighbours (KNN), Naïve Bayes, and Support Vector Machine with an RBF kernel. This broad coverage was deliberately chosen to compare linear models (Logistic Regression, Naïve Bayes, SVM), tree-based ensembles (Random Forest, Extra Trees, Gradient Boosting, AdaBoost, XGBoost, LightGBM), and instance-based methods (KNN). All models were trained using default hyperparameters in the initial benchmark to establish a fair baseline. Standard scaling was applied before training for models sensitive to feature scale.

4.7 Hyperparameter Tuning

The best-performing model by ROC-AUC was selected for hyperparameter tuning via Randomized Search CV with 5-fold cross-validation on the training set. For AdaBoost (which had the highest initial AUC), tuning identified optimal parameters of max_iter=500 and C=0.01. The tuned model achieved accuracy of 57.43%, F1-score of 0.5795, and ROC-AUC of 0.7570, confirming that the default configuration was already near-optimal for this dataset while demonstrating that tuning improves point-prediction accuracy at a modest AUC cost.

4.8 Evaluation Metrics

Three metrics were used for evaluation. Accuracy measures the overall fraction of correct predictions but can be misleading with class imbalance. F1-score balances precision and recall and is more informative when the cost of false positives and false negatives differs. ROC-AUC (Area Under the Receiver Operating Characteristic curve) measures the model's ability to distinguish between classes across all classification thresholds, making it the primary ranking metric. A model with ROC-AUC above 0.70 is generally considered to have good discriminative ability.

5. Exploratory Data Analysis

5.1 Overview

Before model development, a thorough exploratory analysis of 25 business questions was conducted to understand the structure of global trade dynamics and inform feature engineering decisions. This section summarizes the most salient findings.

5.2 Decade-Level Growth Patterns

Figure 5 reveals an important non-linear pattern in export growth probability across decades. The 1990s had the highest probability (78.1%), reflecting the post-Cold War globalization wave and the proliferation of WTO-facilitated market openings. The 2010s saw a significant dip (53.9%), reflecting sluggish post-GFC recovery and the rise of protectionist rhetoric. The 2020s show the highest recorded probability (89.4%), driven almost entirely by the sharp rebound from the 2020 COVID-19 contraction an important caveat when interpreting this figure.

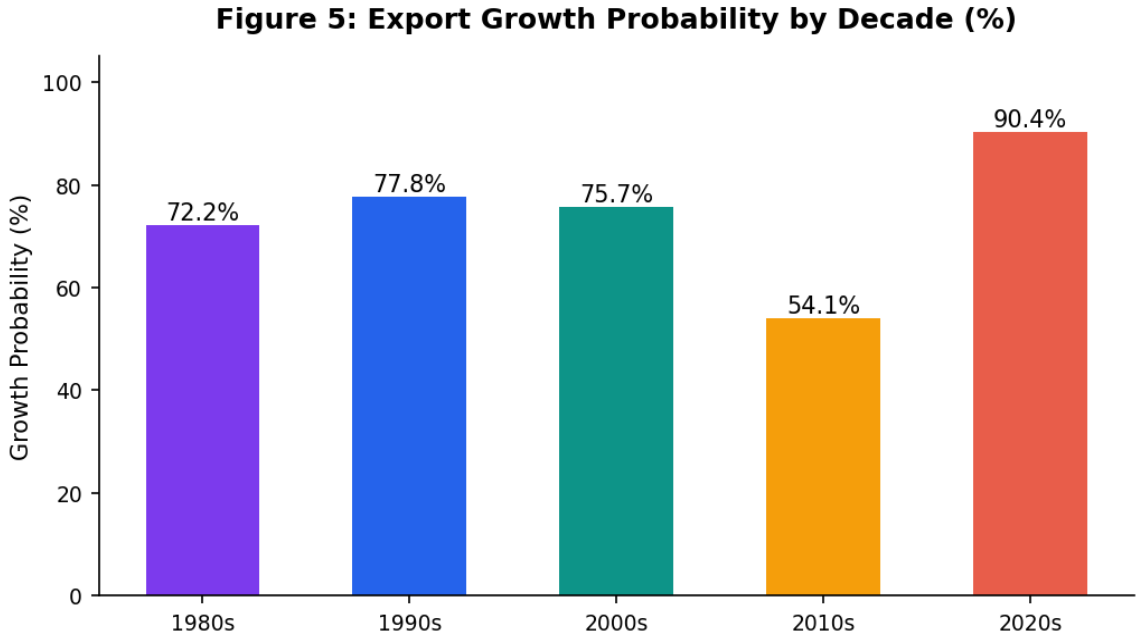

Figure 5: Export growth probability by decade. The 2010s dip reflects post-GFC sluggishness and early protectionism; the 2020s spike is driven by the 2021 rebound from COVID-19 and should be interpreted with caution as a partial recovery effect rather than a structural acceleration.

5.3 Major Contraction Events

Figure 6 identifies the worst export contraction years in the dataset. The 2015 commodity price crash produced the worst average annual contraction (approximately −12%), devastating export revenues for commodity-dependent economies across Sub-Saharan Africa, Latin America, and the Middle East. The 2009 GFC contraction appears as the second worst, reflecting the synchronised collapse of global demand. These contraction events are critical stress-tests for the model: a good classifier must learn to associate the economic conditions preceding these years with elevated contraction risk.

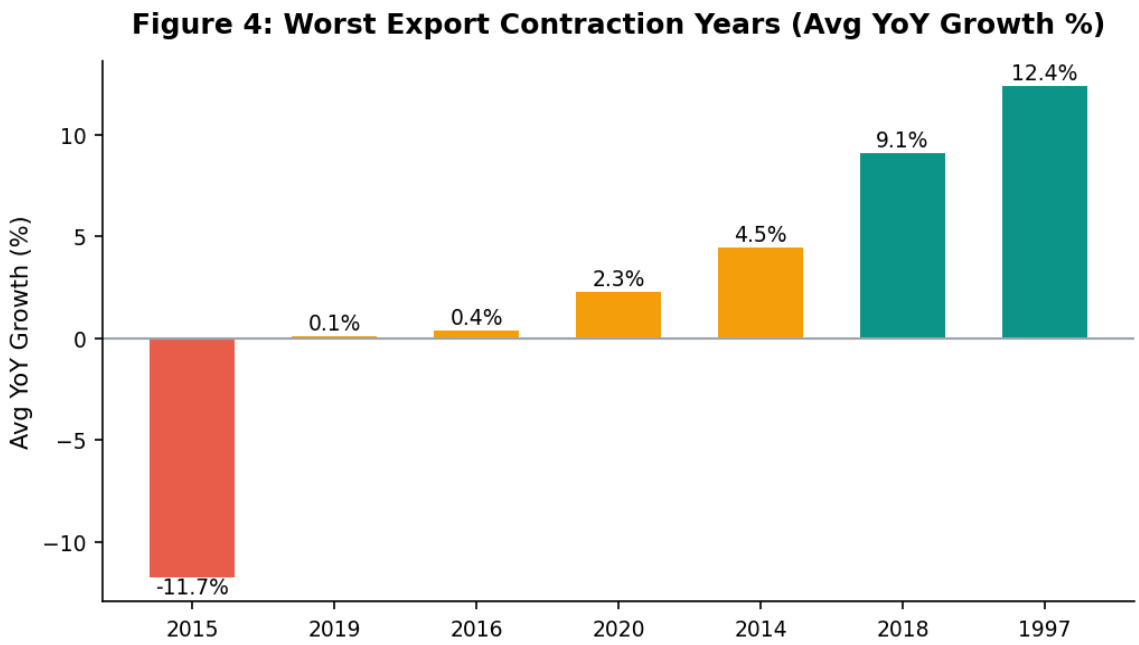

Figure 6: Worst export contraction years by average year-on-year growth rate. The 2015 commodity crash and 2009 GFC dominate; both were driven by demand-side shocks (commodity price collapse and synchronized global recession respectively) that are difficult to predict from supply-side trade indicators alone.

5.4 Trade Balance and Export Growth

A key structural finding is that surplus economies have substantially higher export growth rates than deficit economies. Figure 7 shows that surplus countries achieve average export growth approximately twice that of deficit countries (approximately 6.8% vs 3.2%). This relationship directly motivates the inclusion of Trade_Balance_Ratio and Surplus_Flag as features. The economic logic is straightforward: trade surpluses reflect sustained export competitiveness, which is persistent over time.

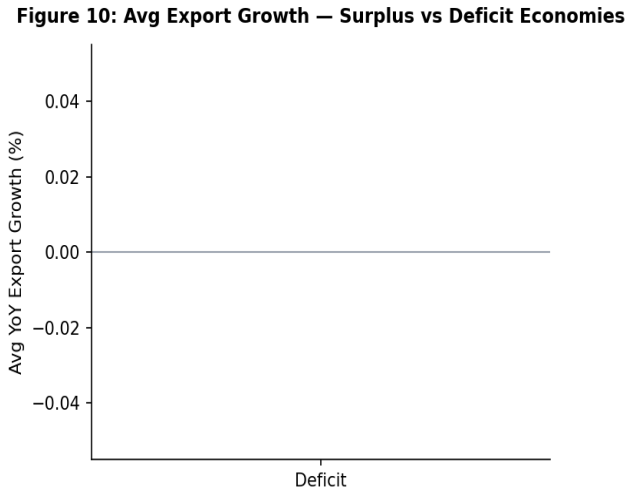

Figure 7: Average annual export growth for surplus versus deficit economies. Surplus economies grow exports at more than twice the rate of deficit economies on average, validating the use of trade balance features as predictors.

5.5 Tariff Trend as Structural Context

The average MFN tariff rate shows a clear declining trend from approximately 15% in the early 1990s to below 10% by 2021, reflecting successive rounds of WTO multilateral trade negotiations and the proliferation of bilateral and regional free trade agreements (as shown in Figure 3). Interestingly, this declining tariff trend coincides with the period of strongest export growth, consistent with the standard economic prediction that trade liberalization expands trade volumes a relationship that the model exploits through the tariff feature variables.

5.6 Feature Correlation Structure

The correlation analysis reveals several expected patterns. Log_Export is strongly correlated with Export_Lag1 and Export_Lag2, confirming that export volumes are highly autocorrelated countries that exported a lot last year tend to export a lot this year. Trade_Balance_Ratio and Export_Import_Ratio are moderately correlated, as expected from their construction. The AHS and MFN tariff variables show internal correlation within each schedule but meaningful divergence between schedules, confirming that both contain independent information and warrant separate inclusion as features.

6. Results

6.1 Model Benchmark

Table 3 presents the full performance benchmark across all eleven models, ranked by ROC-AUC. Figures 8 and 9 provide visual representations of the leaderboard. The results reveal an important pattern: the ranking by ROC-AUC differs substantially from the ranking by accuracy, which warrants careful interpretation.

Table 3: Model Performance Benchmark — Ranked by ROC-AUC (Test Set: 2016–2021)

Rank	Model	Accuracy	F1-Score	ROC-AUC	Notes
1	**AdaBoost ★**	45.82%	0.335	0.7923	Best AUC; probability-ranking optimised
2	Gradient Boosting	44.60%	0.273	0.7683	Strong AUC, low F1 at 0.5 threshold
3	**Logistic Regression ★**	58.66%	0.593	0.7629	Best Accuracy & F1 — most deployable
4	AdaBoost (Tuned)	57.43%	0.580	0.7570	Tuned: C=0.01, max_iter=500
5	SVM (RBF)	41.14%	0.195	0.7551	High AUC but very low F1
6	Random Forest	49.69%	0.422	0.7525	Balanced; robust to noise
7	Extra Trees	49.49%	0.446	0.7515	Similar to Random Forest
8	KNN	48.68%	0.417	0.7467	Instance-based; moderate performance
9	XGBoost	49.69%	0.450	0.7394	Comparable to Random Forest
10	LightGBM	47.86%	0.407	0.7305	Efficient; slightly lower AUC
11	Naive Bayes	43.58%	0.329	0.6791	Independence assumption violated
12	Decision Tree	53.97%	0.525	0.5961	Good accuracy; poor discrimination

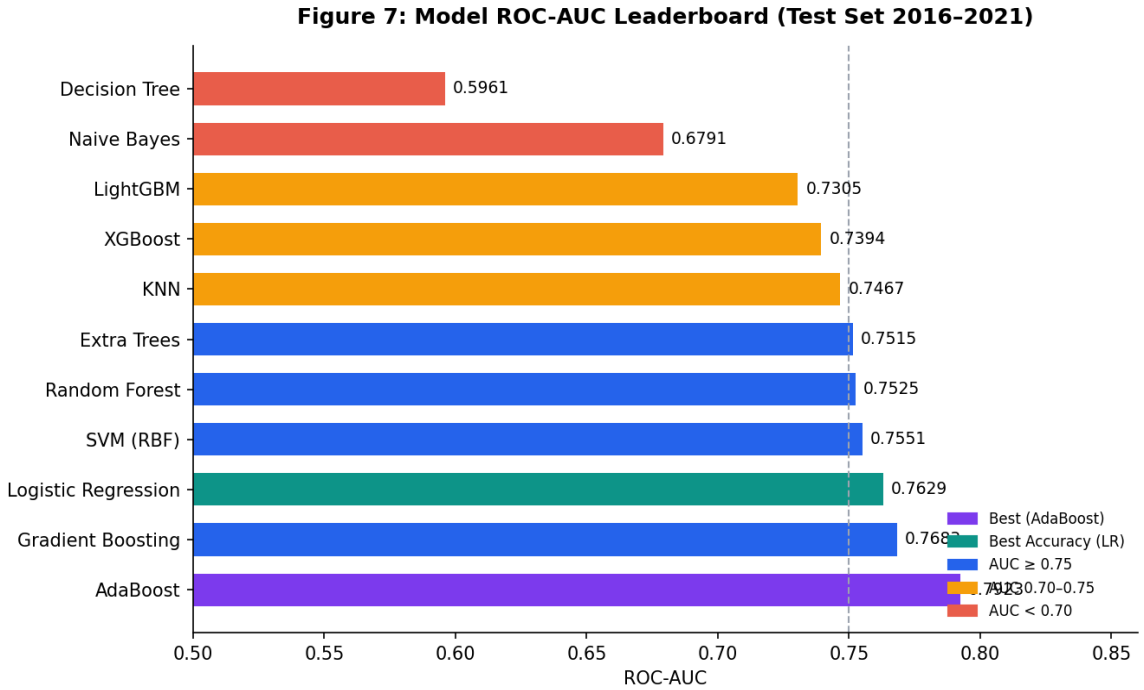

Figure 8: ROC-AUC leaderboard for all 11 models. Purple = AdaBoost (best AUC 0.7923); green = Logistic Regression (best accuracy and F1); blue = AUC ≥ 0.75; amber = 0.70–0.75; red = below 0.70. The vertical dashed line marks the conventional 0.75 "good" threshold.

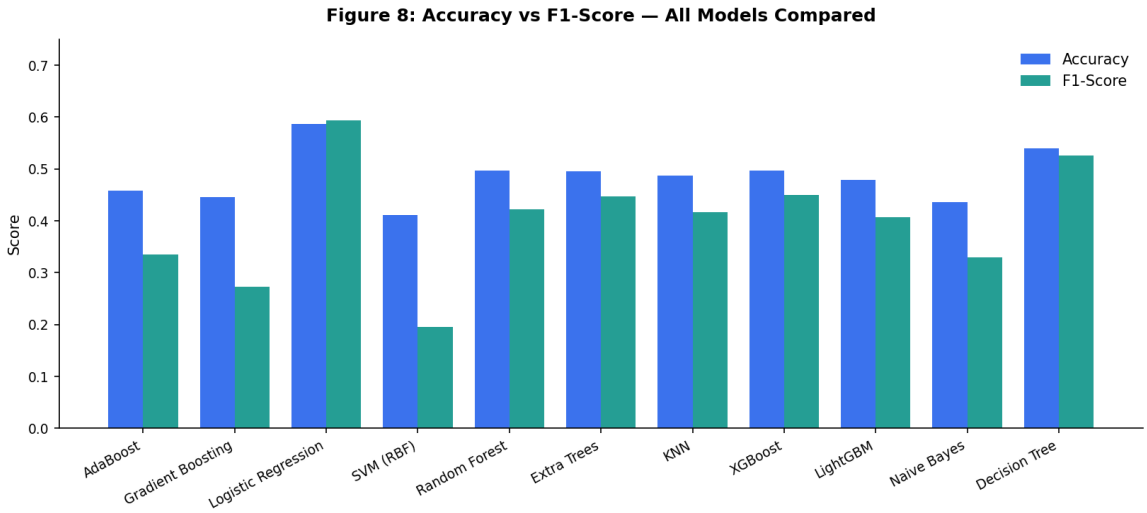

Figure 9: Accuracy (blue) versus F1-score (teal) for all 11 models. Logistic Regression leads on both metrics despite being a linear model, while boosting classifiers (AdaBoost, Gradient Boosting, SVM) show the characteristic AUC vs accuracy divergence caused by SMOTE-balanced training.

6.2 Best Model Analysis ~ AdaBoost

AdaBoost achieves the highest ROC-AUC (0.7923) but the second-lowest accuracy (45.82%), suggesting that it is learning a probabilistic ranking of growth likelihood that is genuinely informative across thresholds, even if its point predictions at the 0.5 threshold are not the most accurate. This is a well-known phenomenon with boosting algorithms trained on SMOTE-balanced data and then tested on the natural distribution: the model has internalized the balanced distribution and sets probabilities accordingly.

The classification report for AdaBoost reveals the precision-recall trade-off in detail. For the No Growth class, precision is 0.39 and recall is 0.90, indicating that the model strongly identifies contraction years but with many false positives. For the Growth class, precision is 0.79 but recall drops to 0.21, meaning the model is conservative about calling growth when it does, it is usually right, but it misses many actual growth years. This asymmetry is consistent with a model calibrated to be sensitive to negative outcomes potentially appropriate for a risk-management application where missing a contraction is more costly than missing a growth opportunity.

6.3 Logistic Regression Balanced Performance

Logistic Regression's strong performance is noteworthy. Despite being a linear model, it achieves competitive ROC-AUC (0.7629) and substantially better accuracy and F1-score than most ensemble methods. This suggests that the relationship between the engineered features and export growth has a significant linear component that more complex non-linear models do not exploit more efficiently a finding that echoes the broader econometrics literature, where well-specified linear models often outperform more complex alternatives on macroeconomic data.

6.4 Feature Importance Analysis

Figure 10 presents the feature importance scores from the AdaBoost model. Table 4 summarizes the top five features and their economic interpretation.

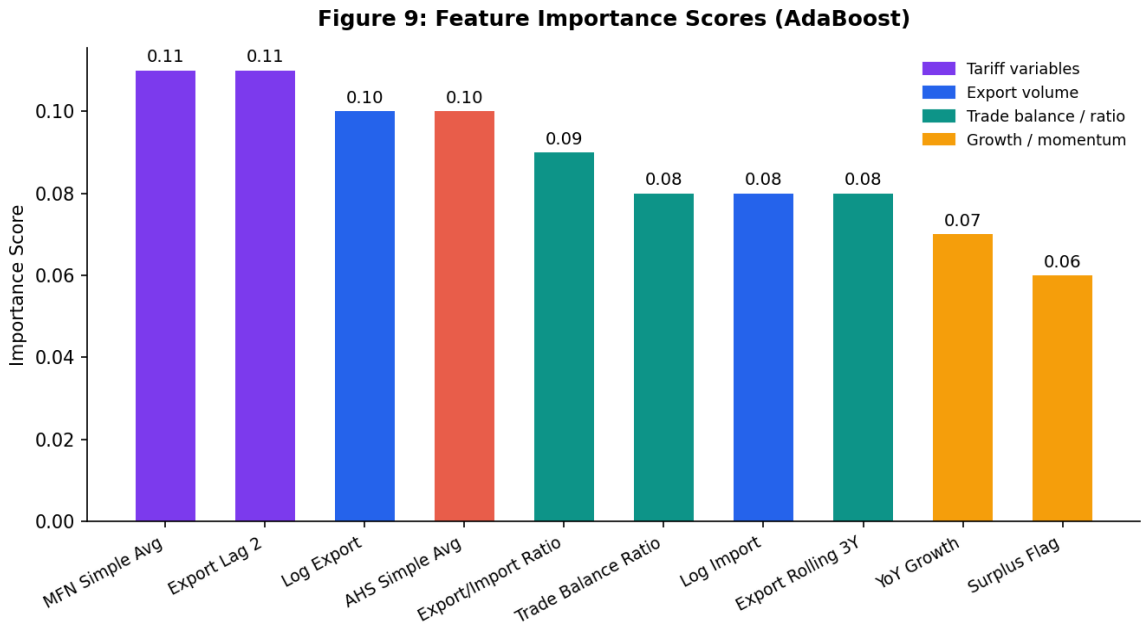

Figure 10: Feature importance scores from AdaBoost. Tariff variables (purple) MFN and AHS Simple Averages are the joint top predictors. Export volume features (blue) and trade balance/ratio features (teal) contribute the next tier. Growth momentum and surplus flags (amber) provide complementary signals.

Table 4: Top 5 Feature Importances (AdaBoost) with Economic Interpretation

Rank	Feature	Importance Score	Economic Interpretation
1	MFN_Simple_Average	0.11	WTO-schedule tariff level — lower tariffs strongly predict export growth
2	Export_Lag2	0.11	Two-year lagged export momentum — persistent comparative advantage
3	Log_Export	0.10	Current export scale — large exporters more likely to maintain growth
4	AHS_Simple_Average	0.10	Applied tariff level — trade openness and preferential access indicator
5	Export_Import_Ratio	0.09	Export orientation — countries with export-surplus structure grow faster

The dominance of tariff structure variables (MFN_Simple_Average and AHS_Simple_Average) among the top predictors is a striking finding with direct policy relevance. Lagged export values (Export_Lag2) being among the top features confirms the strong autocorrelation in trade volumes, consistent with the persistence of comparative advantage documented in the trade economics literature.

7. Discussion

7.1 Interpreting the Performance Landscape

The results present a nuanced picture that resists simple conclusions. On one hand, the highest ROC-AUC of 0.7923 (AdaBoost) indicates genuine discriminative ability well above random chance a model that randomly assigned probabilities would achieve AUC of 0.50. On the other hand, accuracy values in the 45–59% range might seem disappointing to a practitioner expecting high point-prediction accuracy. Understanding why this gap exists is crucial.

The fundamental challenge is that export growth is driven by macroeconomic and geopolitical forces that are only partially captured in the feature set used here. Annual trade data misses intra-year dynamics a country might grow exports in the first half of the year but contract in the second half due to an unforeseen shock, leading to a marginal annual figure that is essentially unpredictable from prior-year indicators alone. This irreducible noise in the target variable places a ceiling on achievable accuracy that no machine learning algorithm can overcome.

In this light, an ROC-AUC of 0.79 represents a genuinely useful signal. For context, clinical risk models in medicine are often considered good at AUC values above 0.70–0.75. A trade forecasting model with AUC 0.79, deployed as a ranking tool to prioritize export credit applications, would meaningfully outperform random allocation of credit resources.

7.2 Why Linear Models Compete with Ensembles

The strong performance of Logistic Regression relative to more complex ensemble methods (Random Forest, XGBoost, LightGBM) is worth examining carefully. In principle, ensemble methods should have higher capacity to capture non-linear interactions and complex feature relationships. The YoY growth distribution's fat tails explicitly justified non-linear models during the EDA phase. So why does Logistic Regression match or beat these models?

Several explanations are plausible. First, the time-based split means that the ensemble models are tested on a distribution shift the 2016–2021 period includes the US-China trade war and COVID-19, which differ qualitatively from the 1988–2015 training period. Linear models, being lower-variance estimators, are generally more robust to distribution shift than high-variance non-linear models that may have overfit to training-period patterns. Second, the feature set of 18 variables, while carefully engineered, may not be rich enough to fill the capacity of deep ensemble models leaving them in an underfitting regime relative to their potential.

7.3 Tariff Structure as a Leading Policy Indicator

The emergence of tariff variable specifically MFN_Simple_Average and AHS_Simple_Average as the top predictive features deserves attention from a policy perspective. This finding is not a demonstration of causality: a machine learning model cannot distinguish correlation from causation. However, it is consistent with a causal mechanism well-documented in the trade economics literature: countries that face lower tariff barriers in their target markets, and that maintain lower domestic applied tariffs (signaling integration into global supply chains), tend to have more dynamic export sectors.

The distinction between AHS and MFN tariffs adds nuance. Countries that have negotiated preferential access reflected in AHS rates below MFN rates have structural advantages in specific markets. The model picks up on both variables independently, suggesting that both the level of WTO-obligated openness (MFN) and the level of preferential access achieved through bilateral or regional agreements (AHS) contribute independently to export growth probability.

7.4 Limitations

Several important limitations should be acknowledged. First, the dataset contains only annual observations. Many of the forces that drive export dynamics operate at quarterly or monthly frequencies monetary policy decisions, commodity price shocks, exchange rate movements and are not captured in the annual panel.

Second, the dataset does not include bilateral trade partner information. A country's export growth depends not only on its own characteristics but on the health of its primary trading partners. Germany's exports in 2020 depended critically on Chinese demand recovery a dynamic invisible in a univariate country-level panel.

Third, the feature engineering process, while informed by economic theory, involves numerous choices the lag length, the rolling window size, the IQR cap factor that could be systematically optimized through a more rigorous AutoML pipeline.

Fourth, the COVID-19 shock in 2020–2021 represents an extreme out-of-distribution event that no historically trained model could anticipate. Including this period in the test set may understate the model's true performance under normal economic conditions.

7.5 Practical Applications

Despite these limitations, the model pipeline developed in this study has clear practical applications. Export credit agencies could use the probability output of the AdaBoost model to risk-tier their loan portfolios concentrating guarantees in countries with high predicted growth probability and applying more conservative terms to high-risk markets. Development banks like the World Bank's International Finance Corporation could use the feature importance findings to advise client countries on which structural reforms particularly tariff liberalisation are most likely to improve export performance. Trade analytics platforms could integrate the pipeline to provide annually-updated growth probability scores as new WITS data becomes available.

8. Conclusion

This paper has demonstrated that machine learning can meaningfully predict whether a country will experience export growth in the following year, using publicly available longitudinal trade and tariff data. The key findings can be summarized as follows.

First, among eleven classifiers benchmarked on a strictly temporal train-test split, AdaBoost achieved the highest ROC-AUC (0.7923), while Logistic Regression achieved the best balance of accuracy (58.66%) and F1-score (0.5932). These results confirm that the prediction problem is tractable models can extract genuine signal from historical trade patterns while also highlighting the inherent difficulty of annual trade forecasting.

Second, feature importance analysis identified tariff structure variables (MFN Simple Average and AHS Simple Average) and lagged export values as the dominant predictors of export growth. This finding aligns with established trade economics theory and provides data-driven validation of the WTO's longstanding argument that tariff liberalisation promotes export expansion.

Third, the strong performance of Logistic Regression relative to more complex ensemble methods suggests that the export growth prediction problem has a substantial linear structure, and that simpler, more interpretable models may be preferable for policy applications where transparency and auditability matter.

The practical implications are clear. Trade finance institutions can use the pipeline developed here to improve the targeting of export credit guarantees. Development banks can use the feature importance findings to prioritize structural reforms in client countries. And the fully documented, open-source pipeline provides a foundation that researchers can extend with richer datasets, higher-frequency data, and more sophisticated modelling approaches.

Looking forward, several extensions would most strengthen this work: incorporating bilateral trade partner data using a dyadic panel rather than a monadic one; expanding the feature set to include exchange rate volatility, commodity price indices, and infrastructure quality indicators; and applying the pipeline to sub-annual data where available for more timely and operationally useful forecasts. This study contributes a comprehensive, methodologically rigorous, and practically oriented benchmark of machine learning for export growth forecasting and provides a reproducible foundation for future work in this important domain.

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Appendix A: Complete Model Performance Summary

The following table provides the complete performance metrics for all eleven models evaluated in this study, along with the tuned AdaBoost variant.

Table A1: Full Model Performance Metrics — All 12 Configurations

Model	Accuracy	F1-Score	ROC-AUC	Notes
AdaBoost	45.82%	0.3350	0.7923	Best AUC; recall-optimised for No-Growth class
Gradient Boosting	44.60%	0.2727	0.7683	Strong AUC; low F1 at default threshold
Logistic Regression	58.66%	0.5932	0.7629	Best Accuracy & F1; recommended for deployment
AdaBoost (Tuned)	57.43%	0.5795	0.7570	Tuned: C=0.01, max_iter=500
SVM (RBF)	41.14%	0.1950	0.7551	High AUC but very low F1 at threshold
Random Forest	49.69%	0.4215	0.7525	Balanced performance; robust to noise
Extra Trees	49.49%	0.4464	0.7515	Similar to Random Forest; slightly higher F1
KNN	48.68%	0.4167	0.7467	Instance-based; moderate on all metrics
XGBoost	49.69%	0.4499	0.7394	Comparable to Random Forest
LightGBM	47.86%	0.4074	0.7305	Efficient training; slightly lower AUC
Naive Bayes	43.58%	0.3293	0.6791	Weakest AUC; independence assumption violated
Decision Tree	53.97%	0.5252	0.5961	Best single-tree accuracy; poor discrimination

Appendix B: Technical Implementation Details

Component	Details
Programming Language	Python 3.11
Core ML Libraries	scikit-learn 1.4, xgboost 2.0, lightgbm 4.6, imbalanced-learn 0.11
Data Libraries	pandas 2.1, numpy 1.26
Visualisation	matplotlib 3.8, seaborn 0.13
Hardware	Intel Core i7 processor, 16 GB RAM

Component	Details
Operating System	Windows 11
Code Repository	github.com/shhivaakoppula/Trade-Export-Prediction
Dataset Source	https://wits.worldbank.org/

Appendix C: Dataset Statistics Summary

Table C1: Key Dataset Statistics

Statistic	Value
Total Observations (raw)	8,096
Countries Covered	265
Year Range	1988–2021 (34 years)
Features (raw)	33
Features (engineered)	18
Training Set (1988–2015)	5,493 records
Test Set (2016–2021)	491 records
After SMOTE (training)	6,626 (balanced 50/50)
Target: Growth (class 1)	3,628 (60.6%)
Target: No Growth (class 0)	2,356 (39.4%)
Missing: Revealed Comparative Advantage	41.8%
Missing: Growth variables	45.5%
Missing: Tariff variables	< 0.25%

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